Native Lefschetz development dataset

Observations only. The native certificate and replay path replaces the Python execution dependency that prevented completion of the previous42-case production attempt. Historical data and the interrupted run remain unchanged.

Native controls	Durable replay	Production boundary
14 tests passed	14 restored tests passed	0 native production cases
17 arithmetic JSON artifacts	17 files byte-identical	No new degree claim

Native Lefschetz development, 2026-10-30; counts are development evidence

Recorded controls.14tests passed: six new native controls and eight inherited donor tests. The new controls check block Hilbert dimensions, degree17 truncation boundary, the explicit8192-term squarefree witness and a coefficient mutation, leading terms for m3–9 and tied-weight negative controls, complete42-key coverage and duplicate/missing rejection, source/base/normalization binding mutations, and retained arithmetic artifacts.

Raw structure. The17JSON arithmetic artifacts contain structure and Hilbert rows, every normalized witness coefficient and squarefree mask, seven leading-term gates, seven negative gates, and source identity. NATIVE_LEFSCHETZ_FORMAT.json explains every file, bit index, variable order and numeric encoding. Binary and source snapshots, logs from every test invocation and all journal controls are retained in the development archive.

Signal handling. Success, deliberate exit7 and direct controller SIGTERM143 tests pass. The active phase records interruption; all unattempted phases retain status125. The first test only terminated the child; it is retained as narrower evidence alongside the corrected direct-controller test. The original production launcher and its incorrect terminal remain immutable with the previously published correction.

S3 replay. The archive was downloaded and its SHA256 checked. Every archived payload passed its manifest before and after replay. The restored exact native test executable passed all14tests; all17recreated arithmetic JSON files match byte-for-byte. These are development checks, not42production certificates or independent mathematical review.

Remaining boundary. No native production case has run; no new FFD,LFD,regularity or elite-base result is claimed. The written Lefschetz/Koszul argument remains conditional and independently unreviewed. Next work is a new frozen protocol with both native binary identities, a qualified EKS image/execution route, then the complete42case native production and fresh/S3replay. The current EKS overview confirms the earlier source probe ended DeadlineExceeded; it supplies no source-access verdict.

Evidence paths. validation-prime-lefschetz-native-development/custody.json; prime-lefschetz-native-development-archives/.artifacts.json; NATIVE_LEFSCHETZ_FORMAT.json. The full Koblitz/general-binary/prime m2–9 fixed-base objective remains active.